INCH-POUND

MIL-C-70678 (AR)
AMENDMENT 2
31 May 1991

SUPERSEDING
AMENDMENT 1
23 February 1990

MILITARY SPECIFICATION

CASE, PROJECTILE BURSTER, XM216 (M216)

This Amendment forms a part of Military Specification MIL-C-70678 (AR), dated 14 January 1988, and is approved for use by the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Title: Add "(M216)" make this change wherever else the nomenclature appears in the specification.

1.1, line 3: Change "XM722" to "M722" and make this change wherever else the nomenclature appears in the specification.

PAGE 4

Add new paragraph 3.7 as follows:

* "3.7 Hydrotest. Prior to air leak testing each burster casing shall be able to withstand the hydrostatic test load specified on dwg. 9390728 without cracking, leakage, rupture or permanent deformation when tested in accordance with 4.5.5.

PAGE 6

* TABLE I; add the following under "Examination for defects ...4.4.2.1":

* "Hydrotest ..10.. - .. 3.7 ..4.5.5".

PAGE 8

* 4.4.2.1; Critical 3, delete "Visual/Gage; 6.2" and substitute "Gage".

AMSC N/A FSC 1310

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* Critical 5, delete "Presence of reverse taper on press fit surface" and substitute, "Diameter of press fit surface increases while traversing from front to rear (open to closed) end of component".

* Critical 7, delete in its entirety.

* Add new Critical "7 .. Hydrotest .. - .. 100% .. 3.7 .. 4.5.5/Gage".

* Add special "a .. Air or hydrotest stamps missing .. - .. 100% .. 3.2 .. Visual".

PAGE 9

* 4.4.2.1, "Drawing Number" block:

Delete "9390278" and substitute "9390728".

PAGE 10

* 4.4.2.1, Add minor 207 as follows:

"207 .. Surface finish inside tube section improper .. 5 .. - .. 3.2 .. Visual 1 /6.2.

* Add Note "1. Randomly selected casings shall be sectioned if a profilometer is unavailable. Any unit which fails the requirement shall cause the lot to be rejected."

PAGE 11

* 4.4.3.2, delete in its entirety and substitute:

"4.4.3.2 Tension test. In order to determine compliance with the mechanical properties requirements, the hardest and softest casings selected from 4.4.3.4 shall be tested in accordance with 4.5.3."

PAGE 12

* 4.5.3, delete in its entirety and substitute:

"4.5.3 Tension test. (see 4.4.3.2) After successfully passing the leak test, the hardest and softest casings selected from 4.4.3.4 shall have two (2) tension test specimens as prescribed by ASTM Method E-8 machined from each casing (four (4) specimens, total). Failure of any specimen to achieve

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the required tensile strength or elongation requirement shall be cause for rejection of the lot represented by that heat of steel. See 6.7."

PAGE 12

* Add 4.5.5 as follows:

"4.5.5 Hydrostatic test. After final machining but prior to the air leak test in 4.5.2, each burster casisng shall be tested using equipment capable of performing the test accurately and consistently. During the test, the pressure will be applied gradually until the required minimum pressure is reached and held for five (5) seconds, minimum. The equipment shall be calibrated prior to the start of each day's operation and at intervals of not over 4 hours of continuous operation. It shall also be calibrated prior to restarting the operation after a layoff of three hours or more. If it has been found that the equipment is out of calibration in such a direction that the pressure was actually lower than required by the test, the casings tested since the last satisfactory calibration shall be retested after the calibration error has been corrected. Casing rupture, cracking, leakage or bulging during the test shall be cause for rejection of the unit. The test fixture shall not restrict the casing from expanding longitudinally or radially under pressure. Protective barriers should be installed to shield operators from metal fragments which may come from defective casings that fail the test. If threads are engaged, the hydrotest plug should be designed with a one quarter inch minimum thread relief, to prevent loading of initial threads on the burster casing. Examination should be made of all hydrotest plugs after a casing is blown during testing. Badly nicked or damaged plugs may cause destruction of acceptable items. After successfully passing the hydrotest but prior to the air leak test, excess water shall be removed from the burster casings and each casing shall be dried at 250 degrees F, minimum for 20 minutes, minimum."

PAGE 13

* Add "6.7 Tension test specimen preparation. After selecting burster casings for tensile testing, it is recommended that the surface finish inside the tube portion of the casings be machined to a final finish of 60 to 125 for accurate test results."

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